

Chapter 80

Rotoscoping with Masks

This chapter covers how to use masks to rotoscope, one of the most common tasks in compositing.

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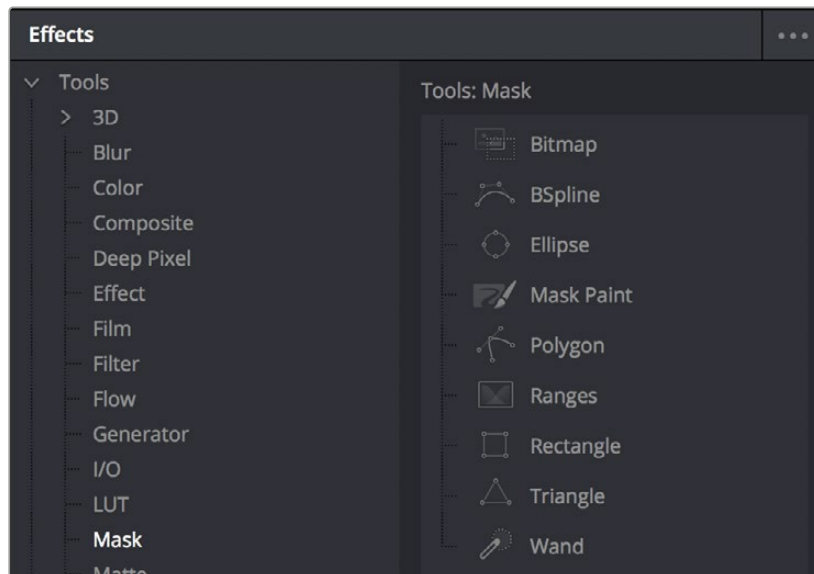
Introduction to Masks and Polylines

Polylines are splines that are used whenever a control is animated with a motion path or when a node's effect is masked with a drawn shape. They are also used in the Paint and Grid Warp nodes. In a more basic form, polylines are used to control the animation in the Spline Editor. Since these splines are used for just about everything, they are extremely flexible, with a considerable amount of controls, modes, and options. This chapter offers an overview of polylines and their operation, with specific information on how to use them for masks.

Mask Nodes

Mask nodes create an image that is used to define transparency in another image. Unlike other image creation nodes in Fusion, mask nodes create a single channel image rather than a full RGBA image. The most used mask tool, the Polygon mask tool, is located in the toolbar.

For more information on these mask tools, see *Chapter 108, "Mask Nodes,"* in the DaVinci Resolve Reference Manual or *Chapter 48* in the Fusion Reference Manual.



The available nodes in the Mask bin of the Effects Library

Polygon Mask

Polygon masks are user-created Bézier shapes. This is the most common type of polyline and the basic workhorse of rotoscoping. Polygon mask tools are automatically set to animate as soon as you add them to the Node Editor.

B-Spline Masks

B-Spline masks are user-created shapes made with polylines that are drawn using the B-Splines. They behave identically to polyline shapes when linear, but when smoothed the control points influence the shape through tension and weight. This generally produces smoother shapes while requiring fewer control points. B-Spline mask tools are automatically set to animate as soon as you add them to the Node Editor.

Bitmap Masks

The Bitmap mask allows images from the Node Editor to act as masks for nodes and effects. Bitmap masks can be based on values from any of the color, alpha, hue, saturation, luminance, and the auxiliary coverage channels of the image. The mask can also be created from the Object or Material ID channels contained in certain 3D-rendered image formats.

Mask Paint

Mask Paint allows a mask to be painted using Fusion's built-in vector paint nodes.

Wand Mask

A Wand mask provides a crosshair that can be positioned in the image. The color of the pixel under the crosshair is used to create a mask, where every contiguous pixel of a similar color is also included in the mask. This type of mask is ideal for isolating color adjustments.

Ellipse, Rectangle, and Triangle Masks

These are primitive shape masks.

For more information on these mask tools, see *Chapter 108, “Mask Nodes,”* in the DaVinci Resolve Reference Manual or Chapter 48 in the Fusion Reference Manual.

Ranges Mask

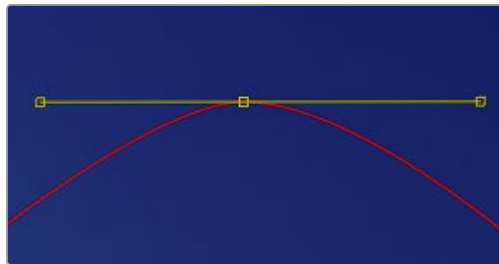
Similar to the Bitmap mask, the Ranges mask allows images from the Node Editor to act as masks for nodes and effects. Instead of creating a simple luminance-based mask from a given channel, Ranges allows spline-based selection of low, mid, and high ranges, similar to the Color Corrector node.

Polyline Types

You can draw polylines using either B-Spline or Bézier spline types. Which you choose depends on the shape you want to make and your comfort with each spline style.

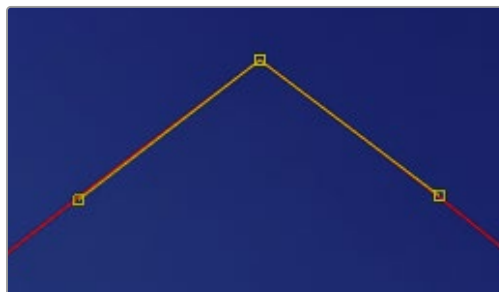
Bézier Polylines

Bézier polylines are shapes composed of control points and handles. Several points together are used to form the overall shape of a polyline.



Bézier control point with direction handles extended to create a smooth curve

Each control point has a pair of handles used to define the exact shape of the polyline segments passing through each control point. Adjusting the angle or length of the direction handles will affect whether that segment of the polyline is smooth or linear.

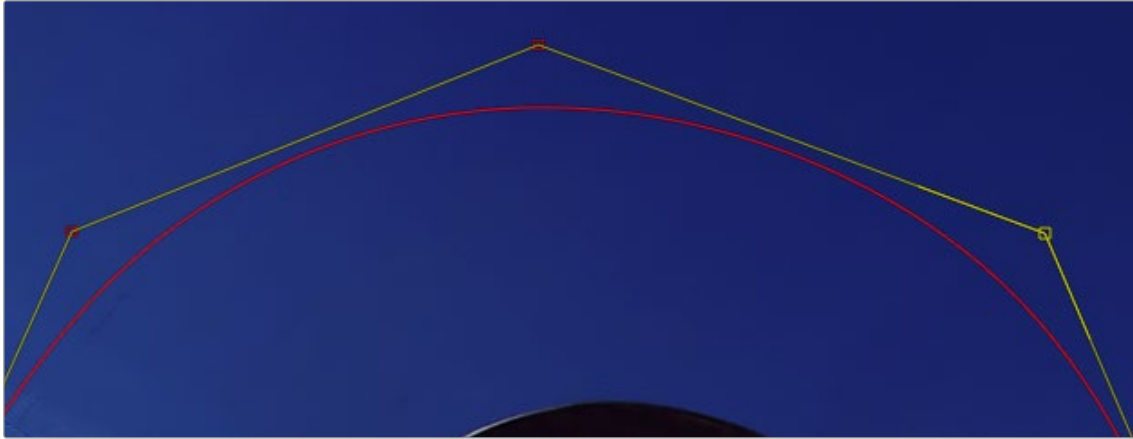


Bézier control point with direction handles aligned to create a linear segment

If you're familiar with applications such as Adobe Photoshop or Illustrator, you'll already be familiar with many of the basic concepts of editing Bézier polylines.

B-Spline Polylines

A B-Spline polyline is similar to a Bézier spline; however, these polylines excel at creating smooth shapes. Instead of using a control point and direction handles for smoothness, the B-Spline polyline uses points without direction handles to define a bounding box for the shape. The smoothness of the polyline is determined by the tension of the point, which can be adjusted as needed.



B-Splines excel at creating smooth curves

Converting Polylines from One Type to Another

Just because you created a shape using a B-Spline or polyline, that doesn't mean you're stuck with the controls you started with. You can convert any shape from B-Spline to Bézier, or Bézier to B-Spline, as needed.

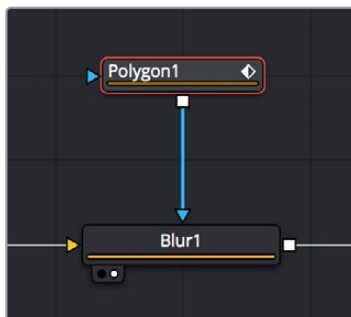
To switch a shape between Polyline and B-Spline controls:

Right-click a shape in the viewer and choose Convert Bézier Spline to B-Spline or Convert B-Spline to Bézier from the spline's contextual menu (only the appropriate option will be displayed).

When converting from one type to another, the original shape is preserved. The new polyline generally has twice as many control points as the original shape to ensure the minimum change to the shape. While animation is also preserved, this conversion process will not always yield perfect results. It's a good idea to review the animation after you convert spline types.

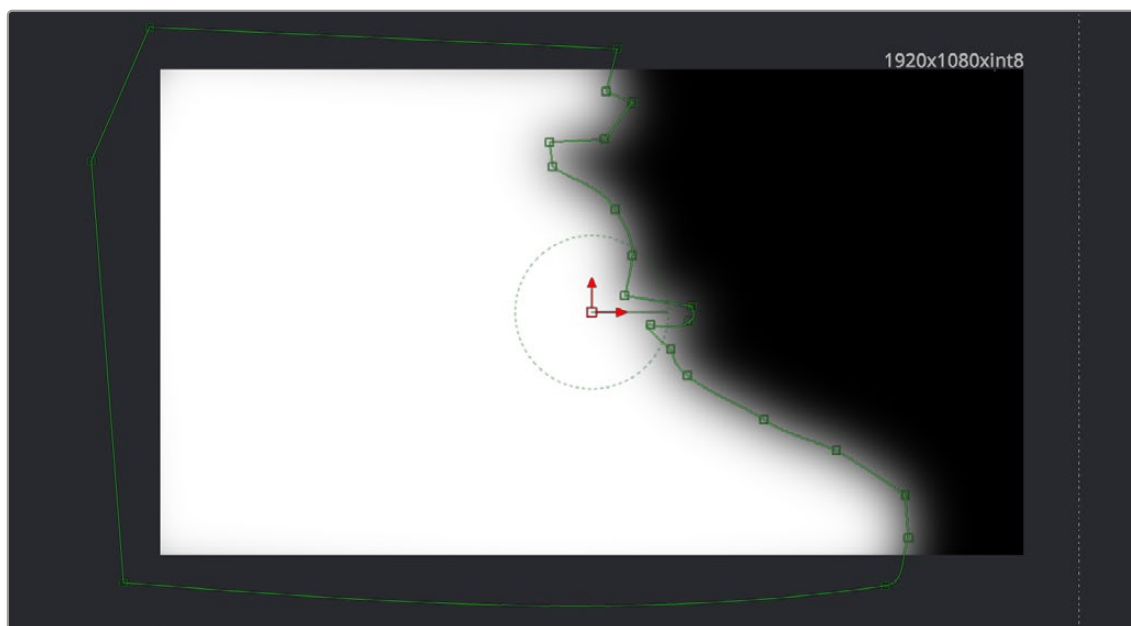
How to Use Masks with Other Nodes

Typically, a node applies its effect to every pixel of an image. However, many nodes have mask inputs that can be used to limit the effect that node has on the image.



A Blur node with a Polygon node masking its effect

Masks are single-channel images that can be used to define which regions of an image you want to affect. Masks can be created using primitive shapes (such as circles and rectangles), complex polyline shapes that are useful for rotoscoping, or by extracting channels from another image.



A Polygon node's mask seen in the viewer

Each mask node is capable of creating a single shape. However, Mask nodes are designed to be added one after the other, so you can combine multiple masks of different kinds to create complex shapes. For example, two masks can be subtracted from a third mask to cut holes into the resulting mask channel.

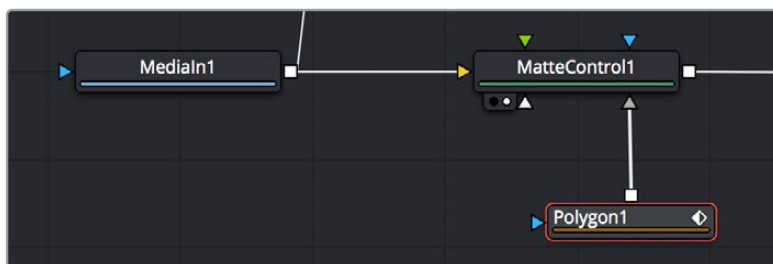
Fusion offers several different ways you can use masks to accomplish different tasks. You can attach Mask nodes after other nodes in which you want to create transparency, or you can attach Mask nodes directly to the specialized inputs of other nodes to limit or create different kinds of effects.

Attaching Masks to an Image for Rotoscoping

There are two ways you'll typically attach a Mask node, such as a Polygon node, so that it adds an alpha channel to an image for compositing later in the node tree.

Using a MatteControl Node

The MatteControl node is the main node used for combining masks in different ways and inserting the result into an image stream. The MatteControl node is attached downstream of the node outputting the image you want to rotoscope. You'll typically attach a Polygon or B-Spline node to the Garbage Matte input of the MatteControl node to use the spline as an alpha channel.



Feeding a Polygon node to a MatteControl node to perform rotoscoping

To use this setup, you'll load the MatteControl node into the viewer and select the Polygon node to expose its controls so you can draw and modify a spline while viewing the image you're rotoscoping. The MatteControl node's Garbage Matte > Invert checkbox lets you choose which part of the image becomes transparent.

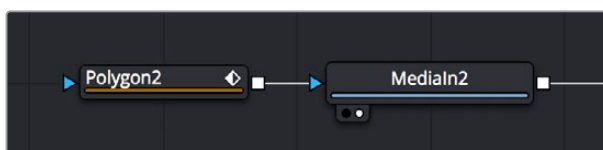
Connecting a Mask to a MediaIn or Loader Node's Input

This method is a bit simpler but requires you to know that you can view one node while adjusting another node, even if that other node is disconnected. If you add an unattached Mask node such as a Polygon or B-Spline node, and then place a MediaIn or Loader node directly into the viewer while selecting the Mask node, you can draw a spline to rotoscope the image.



Rotoscoping a MediaIn node using a disconnected Polygon node

When you're finished rotoscoping, you simply connect the Polygon node's output to the Loader node's input, and an alpha channel is automatically added to that node.

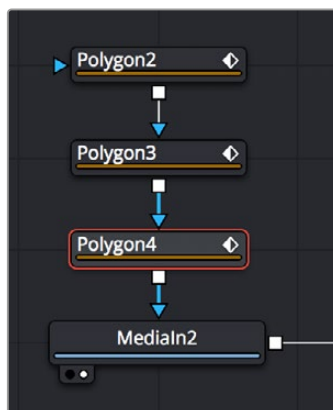


Connecting a Polygon node to a MediaIn node to use a spline as an alpha channel

TIP: If you connect a Mask node to a MediaIn or Loader node's effect input without any shapes drawn, that mask outputs full transparency, so the immediate result is that the image output by the MediaIn or Loader node becomes completely blank. This is why when you want to rotoscope by connecting a mask to the input of a MediaIn or Loader node, you need to work within a disconnected Mask node first. Once the shape you're drawing has been closed, connect the Mask node to the MediaIn or Loader's input, and you're good to go.

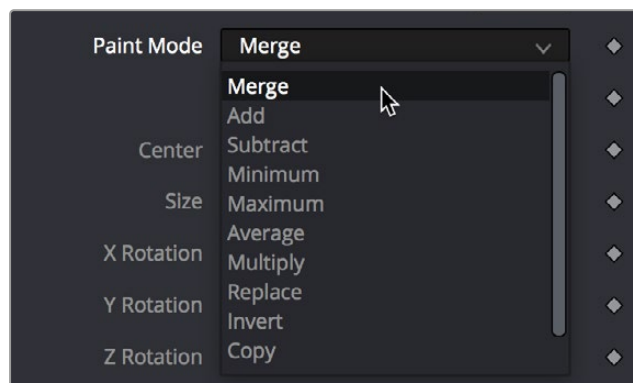
Combining Multiple Masks

Masks are designed to be added one after the other, with each Mask node acting as an additional layer of masking.



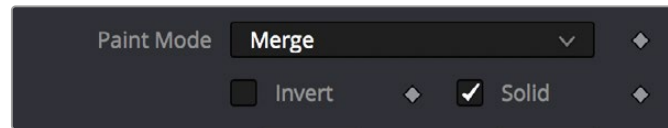
Combining multiple Polygon nodes one after the other in the node tree

When a Mask node's input is attached to another mask, a Paint Mode drop-down menu appears, which allows you to choose how you want to combine the two masks.



The Paint Mode parameter in the Polygon node Inspector parameters

The default option is Merge, but you can also choose Subtract, Minimum, Maximum, Multiply, or any other operation that will give you the mask boolean interaction you need. Additionally, a pair of Invert and Solid checkboxes let you further customize how to combine the current mask with the one before it.



The Invert and Solid options

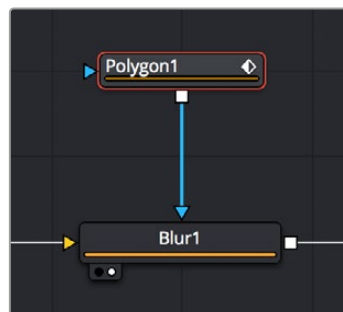
Mask Inputs on Other Nodes

Masks can be used for a variety of reasons, so there are several categories of mask inputs that different nodes include to accommodate these different uses. Incidentally, in most cases you can connect either masks or mattes to a mask input to take advantage of that input's functionality.

TIP: If you select a node with an empty effect mask input, adding a Mask node automatically connects to the open effect mask input.

Effects Mask Inputs

Almost every node in Fusion has an Effect mask input (colored blue), which lets you choose which parts of the image will or will not be affected by that node.

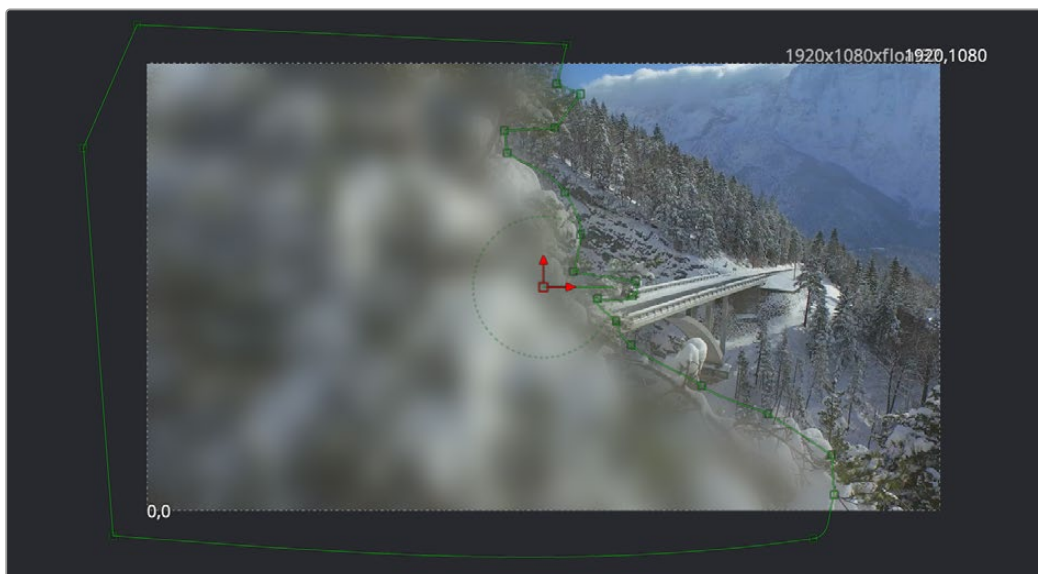


A Blur node with a Polygon node masking its effect

While masks (or mattes) are connected via an input, they are actually applied “post effect,” which means the node first applies its effect to the entire image, and then the mask is used to limit the result by copying over unaffected image data from the input.

Although many nodes support effects masking, there are a few where this type of mask does not apply—notably Savers, Time nodes, and Resize, Scale, and Crop nodes.

TIP: Effects masks define the domain of definition (DoD) for that effect, making it more efficient.



A Polygon node used as a mask to limit the Blur node's effect

Pre-Masking Inputs

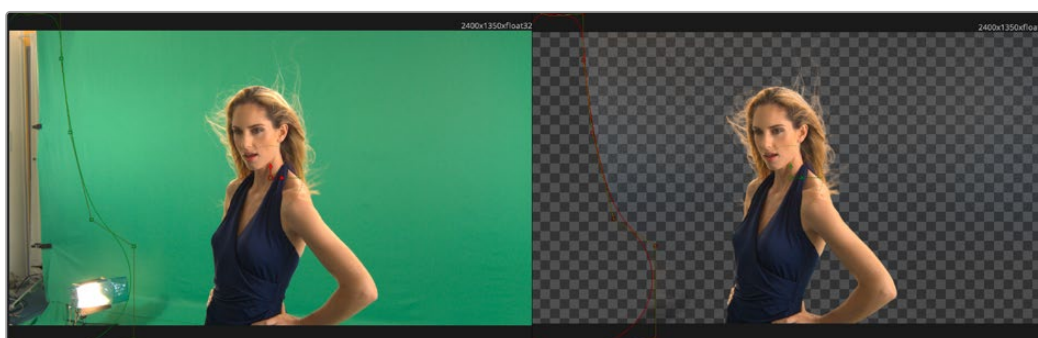
Unlike effect masks, a pre-mask input (the name of which is usually specific to each node using them) is used by the node before the effect is applied. This usually causes the node to render more quickly and to produce a more realistic result. In the case of the Highlight and the Glow nodes, a pre-mask restricts the effect to certain areas of the image but allows the result of that effect to extend beyond the limits of the mask.

The advantage to pre-masking is that the behavior of glows and highlights in the real world can be more closely mimicked. For example, if an actor is filmed in front of a bright light, the light will cause a glow in the camera lens. Because the glow happens in the lens, the luminance of the actor will be affected even though the source of the glow is only from the light.

In the case of the DVE node, a pre-mask is used to apply a transformation to a selected portion of the image, without affecting portions of the image outside of the mask. This is useful for applying transformations to only a region of the image.

Garbage Matte Inputs

Garbage Matte inputs (usually colored gray) are used to exclude light stands, rigging, and boom microphones that intrude upon masks being pulled via blue-screen and green-screen keys. In the following example, a lighting stand to the left is removed from the image via a B-Spline node's mask connected to the Garbage Matte input of the DeltaKeyer node.



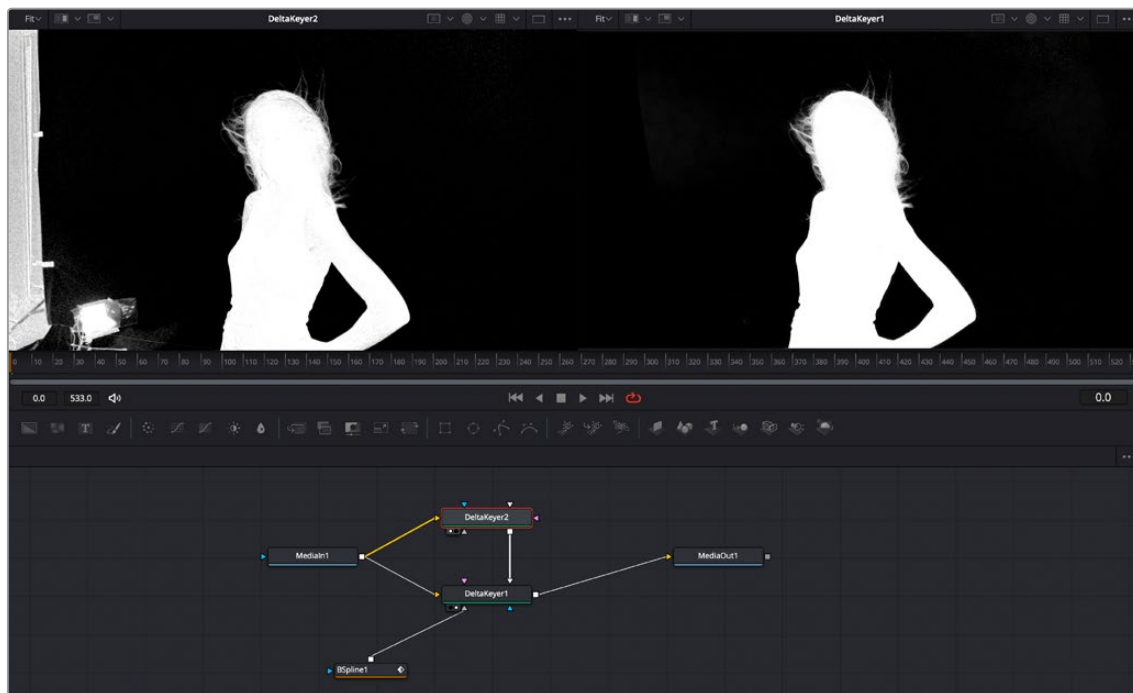
A B-Spline node is connected to the Garbage Matte input of a DeltaKeyer node to eliminate a light stand at the left of frame

TIP: You can quickly add a mask node to the Effect/Solid/Garbage Matte inputs of a keyer node by right-clicking the header bar of that node in the Inspector and choosing whichever mask node you want to use from the Effect Mask, SolidMatte, and GarbageMatte submenus.

You choose whether a garbage matte is applied to a keying node as opaque or transparent in the Inspector for the node to which it's connected.

Solid Matte

Solid Matte inputs (colored white) are intended to fill unwanted holes in a matte, often with a less carefully pulled key producing a dense matte with eroded edges, although you could also use a polygon or mask paint to serve this purpose. In the example below, a gentle key designed to preserve the soft edges of the talent's hair leaves holes in the mask of the woman's face, but using another DeltaKeyer to create a solid matte for the interior of the key that can be eroded to be smaller than the original matte lets you fill the holes while leaving the soft edges alone. This is also sometimes known as a hold-out matte.



Filling in holes in the mask pulled by the DeltaKeyer1 node (left) with another, harder but eroded key in DeltaKeyer2 that's connected to the SolidMatte input of DeltaKeyer1 (right)

Creating and Editing Polylines In-Depth

This section covers the Polygon node's capabilities in depth.

The Polyline Toolbar

Whenever a node that contains one or more polylines is selected, the polyline is shown on all viewers and the Polyline toolbar is displayed along the side of each viewer. The toolbar contains several buttons that make switching polyline modes and options easy to access.



The Polyline toolbar

If you hover the pointer over any of the Polyline toolbar buttons, a tooltip that describes the button's function appears. Clicking on a button will affect the currently active polyline or the selected polyline points, depending on the button.

You can change the size of the toolbar icons, add labels to the buttons, or make other adjustments to the toolbar's appearance in order to make polylines easier to use. All the options can be found by right-clicking on the toolbar and selecting from the options displayed in the contextual menu.

Selecting a Specific Polyline

It is possible to have several polylines in the viewer at once if you select multiple Mask nodes in the Node Editor, so it's important to be able to switch between polylines easily.

To make a polyline active, do one of the following:

- Click one of the polyline's control points or segments.
- Press Tab and Shift-Tab to cycle between available polylines.
- Right-click in the viewer and choose the desired polyline by name from the Controls > Select menu.

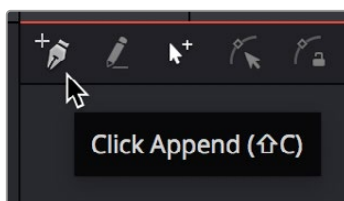
Polyline Creation Modes

There are several different modes available from the toolbar for creating and modifying polylines. The specific mode used when a polyline is first added will depend on whether it is used as a path or a mask.

Each of the modes is described in more detail below.

Click Append

This mode is the default mode for mask creation. It's used to quickly define the rough shape of the mask, after which you switch to Insert and Modify mode to refine the mask further.



The Click Append toolbar button (Shift-C)

To create a mask using the Click Append mode, do the following:

- 1 Select Click Append from the toolbar or press Shift-C.
- 2 Click the pointer where you want to start the shape.
- 3 Move and click the pointer to append a point to the last one.
- 4 To close the shape, place the mouse pointer over the first point created and click when the pointer changes shape.

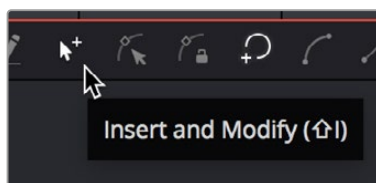
When a shape is closed, the polyline is automatically switched to Insert and Modify mode.

Although the Click Append mode is rarely used with paths, it can be helpful when you know the overall shape of a motion path, but you don't yet know the timing.

TIP: Holding Shift while you draw a mask constrains subsequent points to 45-degree angles relative to the previous point. This can be very helpful when drawing regular geometry.

Insert and Modify

Masks, which are created in Click Append mode, automatically switch to Insert and Modify mode when the mask shape is closed. You can also manually switch to this mode by clicking the Insert and Modify button in the toolbar or using the Shift-I keyboard shortcut. This mode makes it easier to add additional points and refine the shape of the mask. Dragging the control points or direction handles modifies existing points on the polyline.

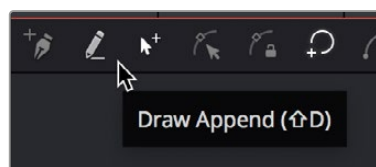


The Insert Modify toolbar button (Shift-I)

Insert and Modify mode is also the default mode for creating motion paths. A new control point is automatically added to the end of the polyline, extending or refining the path, any time a parameter that is animated with a motion path is moved.

Draw Append

The Draw Append mode creates a freehand polyline shape directly on the viewer, like drawing with a pencil or a paintbrush. This mode is ideal to use in conjunction with a tablet and for the creation of garbage mattes and effect masks around complex shapes.



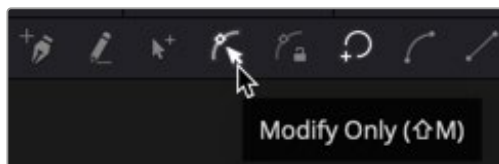
The Draw Append toolbar button (Shift-D)

Protection Modes

In addition to the modes used to create a polyline, two other modes are used to protect the points from further changes after they have been created.

Modify Only

Modify Only mode allows existing points on the polyline to be modified, but new points may not be added to the shape.

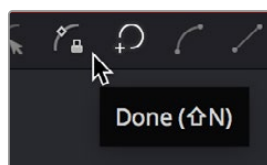


The Modify Only toolbar button (Shift-M)

TIP: Even with Modify Only selected, it is still possible to delete points from a polyline.

Done

The Done mode prohibits the creation of any new points, as well as further modification of any existing points on the polyline.



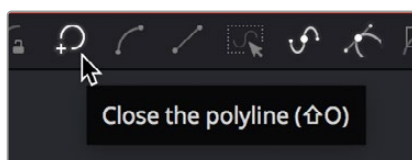
The Done toolbar button (Shift-N)

Closing Polylines

There are several ways to close a polyline, which will connect the last point to the first.

To close a polyline, do one of the following:

- Hover the pointer over the first point created, and then click on the point.
- Press Shift-O on the keyboard.
- Click the Close button on the polyline toolbar.
- Draw a polyline until you are ready to close the shape, and then right-click and choose Polygon:Polyline > Closed.



The Close toolbar button (Shift-O)

All these options are toggles that can also be used to open a closed polygon.

Selecting and Adjusting Polylines

To create the shape you need for a mask or a motion path, you need to know how to manipulate the splines. Fusion provides a number of simple techniques for selecting, moving, and smoothing a spline, but also includes more complex adjustment techniques for scale, skewing, and twisting a spline.

Polyline Points Selection

To select one or more control points on a polyline, do one of the following:

- Click directly on the control points.
- Lasso around the points.

To add or remove points from the current selection, do one of the following:

- Hold the Shift key to select a continuous range of points.
- Hold Command and click each control point you want to add or remove.
- Press Command-A to select all the points on the active polyline.

TIP: Once a control point is selected, you can press Page Down or Page Up on the keyboard to select the next control point in a clockwise or counterclockwise rotation. This can be very helpful when control points are very close to each other.

Moving Polyline Points

The selected polyline points can be moved using either the keyboard or the mouse.

To move selected control points using the pointer, do one of the following:

- Drag on the selected points anywhere in the viewer.
- Hold Shift while dragging to restrict movement to a single axis.
- Hold Option and drag anywhere in the viewer to move the selected control point.

To move selected control points using the keyboard, do one of the following:

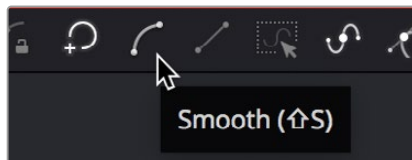
- Press the Up or Down Arrow keys on the keyboard to nudge a point up or down in the viewer.
- Hold Command-Up or Down Arrow keys to move in smaller increments.
- Hold Shift-Up or Down Arrow keys to move in larger increments.

Smoothing a Polyline Segment

If you want to shape the polyline and control its slope, you can choose to smooth a spline segment by adjusting the Bézier direction handles.

To smooth the selected points on an active polyline, do one of the following:

- Press Shift-S.
- Click the Smooth button on the Polyline toolbar.
- Choose Smooth from the polyline's contextual menu.



The Smooth button in the toolbar (Shift-S)

Linearizing a Polyline Segment

To make certain that a polyline segment is perfectly straight, that segment must be linearized. A linear segment aligns the Bézier direction handles with the segment and therefore has no curvatures. The segment is always drawn in a straight line between two points on the polyline.

To linearize the selected points on an active polyline, do one of the following:

- Press Shift-L.
- Click the Linear button on the polyline's toolbar.
- Choose Linear from the polyline's contextual menu.



The Linear button in the toolbar (Shift-L)

Transforming Individual or Multiple Points

Select the points to be transformed, and then do one of the following:

- Hold T and drag to twist.
- Hold S and drag to scale.
- Hold X and drag to scale horizontally only.
- Hold Y and drag to scale vertically only.
- Hold O and drag to offset the points perpendicular to the tangent.

The position of the pointer when the transformation begins becomes the center used for the transformation.

Deleting Selected Points

You can delete a selected point or group of points by pressing Delete or Backspace, choosing Delete from the contextual menu, or by clicking the Delete Point button in the toolbar. The shape of the polyline changes to reflect the removal of these points.

TIP: Deleting all the points in a polyline does not delete the polyline itself. To delete a polyline, you must delete the node or modifier that created the polyline.

Editing Bézier Handles

For Bézier polylines, each control point has two direction handles that adjust the slope of a curve through the control point. These direction handles appear only when the point is selected.

Dragging a direction handle makes adjustments to the curve of the segment that emerges from the control point. The direction handle on the opposing side of the control point will also move to maintain the relationship between these two handles.

To break the relationship between direction handles and adjust one independently, hold Command while dragging a handle. Subsequent changes will maintain the relationship, unless Command is held during each adjustment.

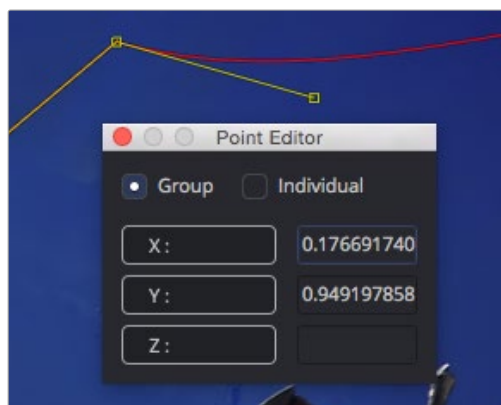


Hold Command to adjust one handle independently

If you want to adjust the length of a handle without changing the angle, hold Shift while moving a direction handle.

Point Editor

The Point Editor dialog can be used to reposition control points using precise X and Y coordinates. Pressing the E key on the keyboard will bring up the Point Editor dialog and allow you to reposition one or more selected control points.

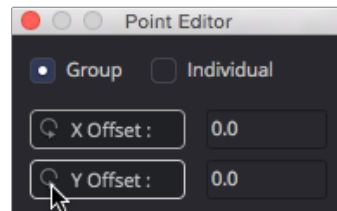


The Point Editor dialog can be used to position control points

The dialog box contains the X- and Y-axis values for that point. Entering new values in those boxes repositions the control point. When multiple control points are selected, all the points move to the same position. This is useful for aligning control points along the X- or Y-axis.

If more than one point is selected, a pair of radio buttons at the top of the dialog box determines whether adjustments are made to all selected points or to just one. If the Individual option is selected, the affected point is displayed in the viewer with a larger box. If the selected point is incorrect, you can use the Next and Previous buttons that appear at the bottom of the dialog to change the selection.

In addition to absolute values for the X- and Y-axis, you can adjust points using relative values from their current position. Clicking once on the label for the axis will change the value to an offset value. The label will change from X to X-offset or from Y to Y-offset.

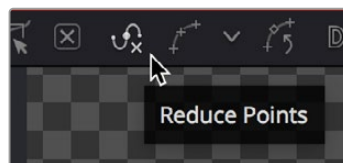


The Point Editor dialog with Offset values

If you are not sure of the exact value, you can also perform mathematical equations in the dialog box. For example, typing 1.0-5 will move the point to 0.5 along the given axis.

Reduce Points

When freehand drawing a polyline or an editable paint stroke, the spline is often created using more control points than you need to efficiently make the shape. If you choose Reduce Points from the polyline's contextual menu or toolbar, a dialog box will open allowing you to decrease the number of points used to create the polyline.



The Reduce Points button in the toolbar

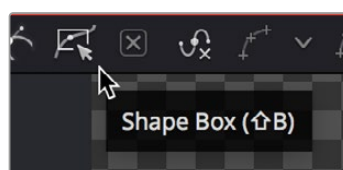
The overall shape will be maintained while eliminating redundant control points from the path. When the value is 100, no points are removed from the spline. As you drag the slider to the left, you reduce the number of points in the path.

Shape Box

If you have a polyline shape or a group of control points you want to scale, stretch, squish, skew, or move, you can use the shape box to easily perform these operations.

To enable the shape box, do one of the following:

- Click the Shape Box toolbar button.
- Choose Shape Box from the contextual menu.
- Press Shift-B.



The Shape Box button in the Polyline toolbar

If there are selected points on the polyline when the Shape Box mode is enabled, the shape box is drawn around those points. Otherwise, you can drag the shape box around the area of control points you want to include.

If you want to freely resize the shape box horizontally and vertically, you can drag a corner handle. Dragging a handle on the side of the shape box resizes the polyline along a specific axis.



Dragging a side handle resizes along a specific axis

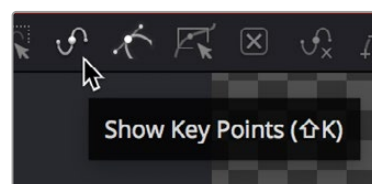
Holding Command while dragging a shape box handle will apply adjustments from the center of the shape box, constraining the transformation to the existing proportions of the shape box. Holding Shift while dragging a corner handle affects only that handle, allowing skewed and non-uniform transformations.



Hold Shift while dragging a corner to perform non-uniform transformations

Showing and Hiding Onscreen Polyline Controls

It is often difficult to identify individual points when they are placed closely together. You can choose to display both points and their direction handles, just points, or just handles. These display mode options are selected using the Show Key Points and Show Handles toolbar buttons, or from the polyline's context menu.



The Show Key Points and Show Handles buttons in the toolbar

You use these options to simplify the screen display when adjusting control points placed closely together and to avoid accidentally modifying controls and handles that are adjacent to the intended target.

Stop Rendering

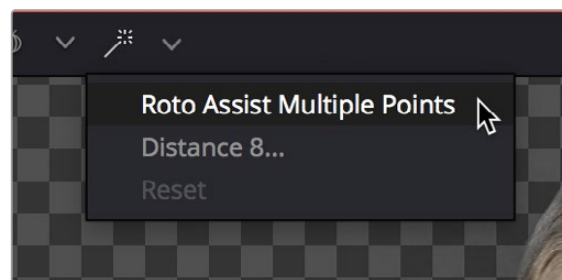
While points along the polyline are being moved, the results are rendered to the viewer to provide constant interactive feedback. Although extremely useful, there are situations where this can be distracting and can slow down performance on a complex effect. To disable this behavior so renders happen only when the points stop moving, you can toggle the Stop Rendering button in the toolbar or select this option from the polyline contextual menu.

Roto Assist

You can enable the Roto Assist button in the toolbar when you begin drawing your shape to have points snap to the closest high-contrast edge as you draw the shape. The points that have snapped to an edge are indicated by a cyan outline.

There are three main Roto Assist options.

- **Multiple Points:** Allows adding multiple points along an entire edge with a single click instead of having to add each point individually.
- **Distance:** Defines the pixel range within which searching for an edge will take place.
- **Reset:** Used for resetting the snap attribute of the snapped points. After resetting, the points will become unavailable for tracking.

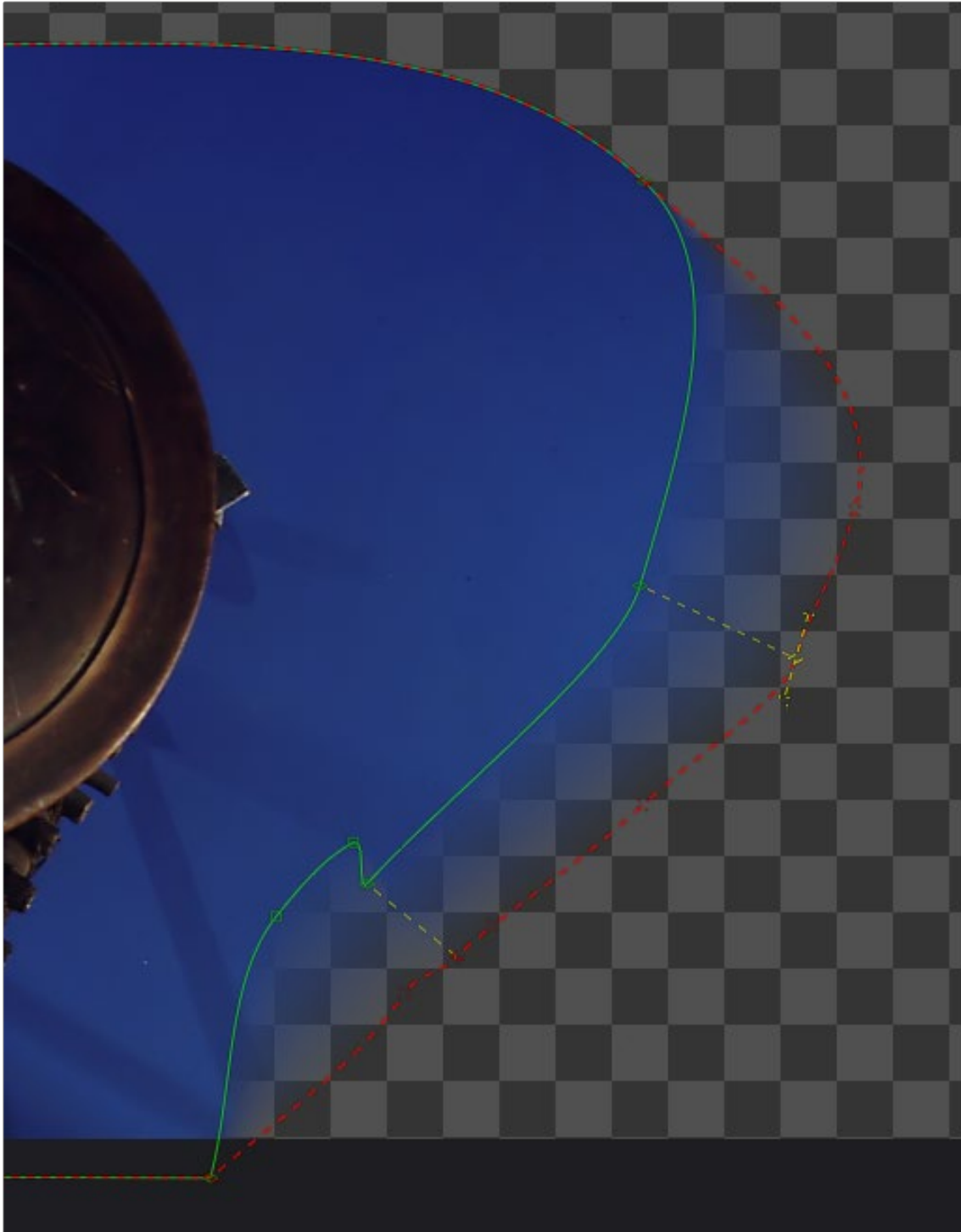


The Roto Assist options in the toolbar

Creating Softness Using Double Polylines

The standard soft edge control available in all Mask nodes softens the entire mask equally. However, there are times, particularly with a lot of motion blur, when softening part of the curve while keeping other portions of the curve sharp is required.

This form of softness is called non-uniform softness, which is accomplished by converting the shape from a single polyline to a double polyline. The double polyline is composed of two shapes: an inner and an outer shape. The inner shape is the original shape from the single polyline, whereas the outer shape is used to determine the spread of the softness. The further the outer shape gets from the inner shape, the softer that segment of the shape becomes.

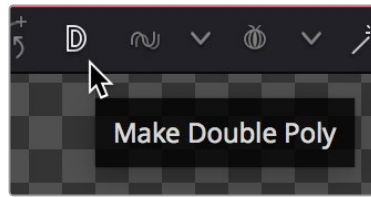


A double polyline uses an inner and outer shape for non-uniform softness

Converting a Single Polyline to a Double Polyline

To convert a mask into a double polyline, click the Double Polyline button in the Polyline toolbar or right-click in the viewer and select Make Outer Polyline from the mask's contextual menu.

The shape will be converted into an inner and an outer polyline spline. Both polylines start with exactly the same shape as the original single polyline. This keeps the mask sharp to start with and allows any animation that may have already been applied to the shape to remain.



Make Double Polyline button

The control points on the outer shape are automatically parented to their matching points on the inner shape. This means that any changes made to the inner shape will also be made to the outer shape. The relationship is one-way; adjustments to the outer shape can be made without affecting the inner shape.

A dashed line drawn between the points indicates the relationship between the points on the inner and outer shapes.

Adding Softness to a Segment

The outer shape is drawn using a green dashed line instead of a solid line to help distinguish it from the inner shape.

To select the outer soft edge shape, do one of the following:

- Use the Tab key to cycle between the onscreen controls until the dashed outline is visible
- Right-click over a spline in the view and choose Controls > Select > Polygon: Outer Polygon.

Once the outer polyline is selected, you can drag any of the points away from the inner polyline to add some softness to the mask.

TIP: Press Shift-A to select all the points on a shape, and then hold O and drag to offset the points from the inner shape. This gives you a starting point to edit the falloff.

The farther the outer shape segment is from the inner shape, the larger the falloff will be in that area.

Adding Additional Points to the Shape

It is not necessary for every point on the inner shape to have a match on the outer shape, or vice versa. You can add additional control points to refine the shape of either shape.

Each polyline stores its animation separately; however, if a point is adjusted on the inner shape that is parented to a point on the outer shape, a keyframe will be set for both splines. Adjusting a parented point on the outer shape only sets a keyframe for the outer shape's spline. If a point that is not parented is adjusted, it will only set a keyframe on the relevant spline. You can disable this behavior entirely for this polyline by selecting Polygon: Outer Polygon > Follow Inner Polyline from the contextual menu.

Locking/Unlocking Point Pairs

If you want to parent additional control points, you can select the points, right-click in the viewer, and choose Lock Point Pairs from the contextual menu for either spline. This will cause the selected point on the outer shape to become parented to the selected point on the inner shape.

Any animation already applied to either point is preserved when the points become parented.

To unlock a point so it is no longer parented, select the point, right-click in the viewer, and deselect Lock Point Pairs from the contextual menu.

Animating Polyline Masks

Animating masks is surprisingly easy. When Polygon or B-Spline masks are added to the Node Editor, the spline's control points are automatically ready to be animated. All you have to do to animate a mask is move the playhead to a new frame and then change the shape of the mask. A new keyframe is added in the Spline Editor and Timeline Editor. This one keyframe controls the position of all control points for that mask at that frame. Once two or more keyframes have been created, the shape of the polygon or B-Spline is automatically interpolated from one keyframe to the next.

TIP: The center point and rotation of a shape are not auto-animated. Only the control points are automatically animated. To animate the center position or rotation, enable keyframes for that parameter in the Inspector.

To adjust the overall timing of the mask animation, you edit the Keyframe horizontal position spline using the Spline Editor or Timeline Editor. Additional points can be added to the mask at any point to refine the shape as areas of the image become more detailed.

Removing Animation from a Polyline Mask

If you want a Polyline mask to remain static, you can remove the automatic animation setting. In the Inspector for the mask, right-click in the bottom of the panel where it says Right Click Here For Shape Animation. From the contextual menu, choose Remove Bézier Spline. If you decide you need to animate the mask at a later time, right-click in the same area again and choose Animate.

Adding and Removing Points from an Animated Mask

When adding points to an animated mask, the new point is fit into the shape at all keyframes. Deleting a point removes that point from all keyframes in the animated mask.

Publishing Specific Control Points

Although you can rapidly animate the entire shape of a polyline using a single keyframe, by default the Spline Editor and Timeline display only one keyframe for the entire shape at any given frame.

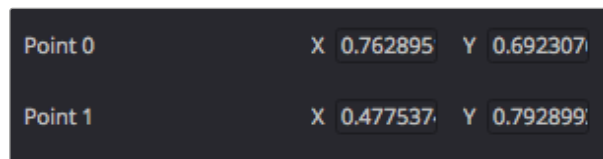
This default keyframing behavior is convenient when quickly animating shapes from one form to another, but it doesn't allow for specific individual control points that need to be keyframed independently of all other control points for a particular shape. If you're working on a complex mask that would benefit from more precise timing or interpolation of individual control points, you can expose one or more specific control points on a polyline by publishing them.

Be aware that publishing a control point on a polyline removes that point from the standard animation spline. From that point forward, that control point can only be animated via its own keyframes on its own animation spline. Once removed, this point will not be connected to paths, modifiers, expressions, or trackers that are connected to the main polyline spline.

To publish a selected point or points, do one of the following:

- Click on the Publish Points button in the Polyline toolbar.
- Select Publish Points from the Polyline's contextual menu.

A new coordinate control is added to the Polyline mask controls for each published point, named Point 0, Point 1, and so on.



The Publish Points controls in the Inspector

The onscreen control indicates published points on the polyline by drawing that control point much larger. Once a published point is created, it can be connected to a tracker, path, expression, or modifier by right-clicking on this control and selecting the desired option from the point's contextual menu.



The published point in the viewer

Using “Publish to Path” to Preserve Animation

When a point is published, any animation already applied to that point is removed. However, if you need to keep the animation, you can use the “Publish to Path” option. This Polyline contextual menu option publishes the selected points and converts their existing animation to a path. You can also use the Publish to Path button in the Polyline toolbar.

Using “Follow Published Points” to Add Points

There are times when you will need to have control points that lie between two other published points follow the motion of the published points, while still maintaining their relative offset and shape. For this reason, points in a Polyline mask can be set to “Follow Published Points” using the Polyline's contextual menu.

When a point of an effect mask is set to follow points, the point will be drawn as a diamond shape rather than a small box.



A control point set to Follow Published Points

When this mode is enabled, the new “following” control points will maintain their position relative to the motion of any published points in the mask, while attempting to maintain the shape of that segment of the mask. Unlike published points, the position of the following points can still be animated to allow for morphing of that segment's shape over time.